COMPREHENSIVE EXPANSION REPORT

Technical Architecture & Operational Rigor Document

Addressing Professor Inquiries: Multi-Process Workflows, Agent-Based Interactions, Sequential Multi-Objective RL, and Causal Feedback Dynamics

COURSE
Simulation for Applied AI

INSTITUTION
Miami Dade College

TEAM MEMBERS
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PRIMARY FOCUS
Option 2 (Campus Rover Logistics) & Option 1

Executive Summary & Response to Faculty Guidance

We sincerely appreciate the professor's encouraging and insightful feedback on our preliminary submission. In particular, we welcome your strong recommendation regarding **Option 2: Smart Campus Autonomous Courier Rover Logistics for Miami Dade College Facilities**.

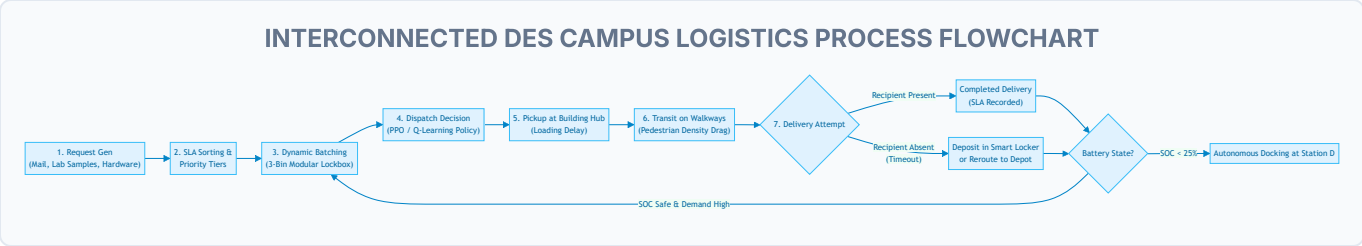
Below, our team provides the rigorous operational processes, agent definitions, reinforcement learning formulation, and causal feedback loops requested. Both options are expanded, with profound emphasis on Option 2 to confirm that our 4-student team has distinct, substantial technical responsibilities spanning Discrete-Event Modeling, Agent-Based interactions, Reinforcement Learning, and interactive dashboard analytics.

SECTION 1: OPTION 2 (RECOMMENDED) — MDC CAMPUS AUTONOMOUS ROVER LOGISTICS

> **Faculty Endorsement:** "I particularly really, really like option 2 because it offers opportunities to investigate operational efficiency, resource allocation, autonomous decision-making, and cost optimization within a realistic campus environment."

1.1 Interconnected Operational Processes (Lifecycle from Request to Delivery)

Rather than a simplistic single queue-and-service model, Option 2 models **seven interconnected operational subsystems** that interact stochastically:

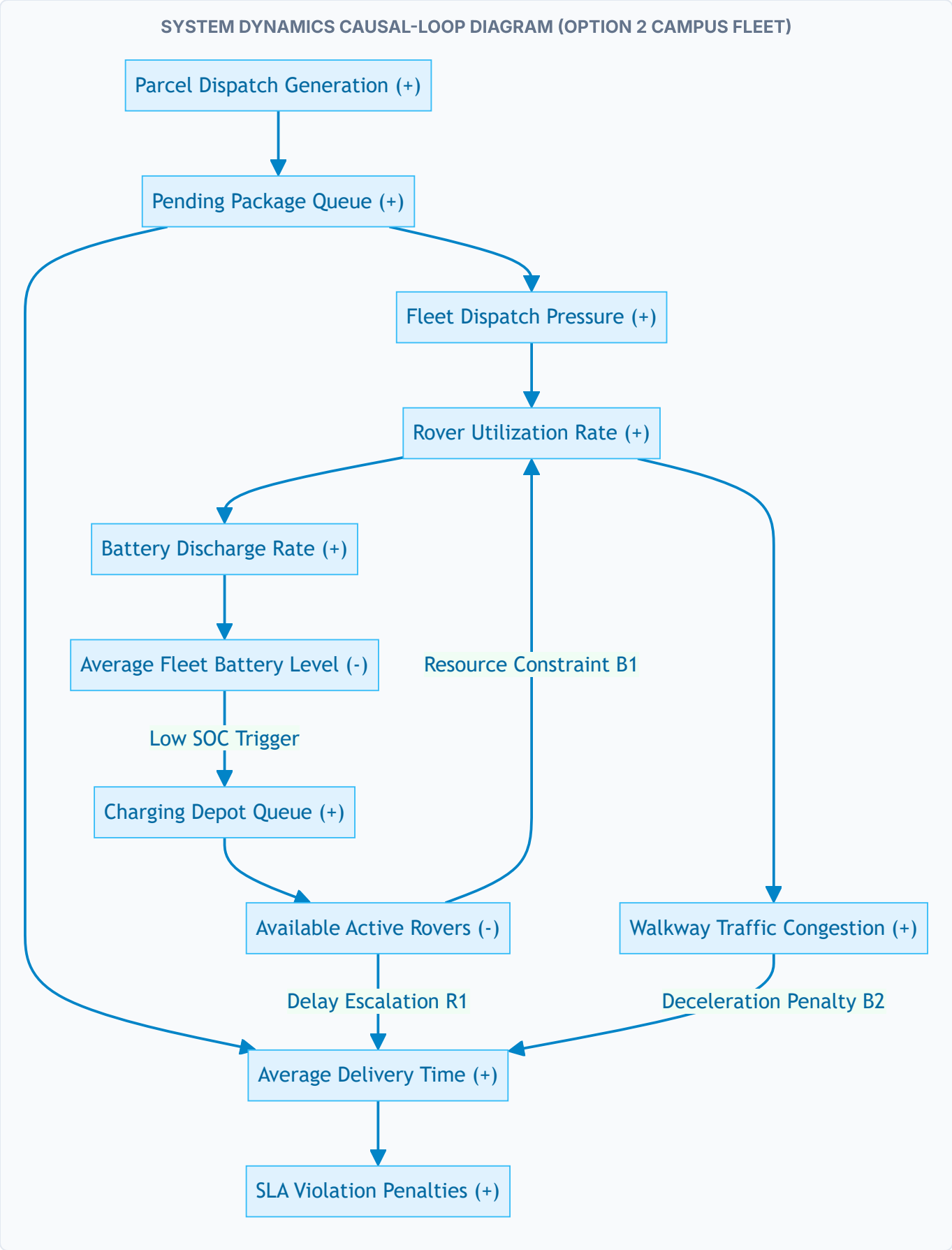


1. **Request Generation Subsystem:** Dynamic arrival rates driven by campus schedule phases (morning administrative mail bursts, midday student lab sample transfers, afternoon urgent department dispatches). Models Poisson arrivals with time-dependent arrival rates λ(t).
2. **Priority & Service Level Agreement (SLA) Stratification:** Requests are segregated into three tiers:
   - **Urgent (Tier 1):** Biohazard/chemistry lab specimens and emergency administrative docs (Strict 15-minute SLA limit).
   - **Standard (Tier 2):** Inter-departmental packages and student equipment transfers (45-minute SLA).
   - **Bulk (Tier 3):** Routine campus mail and recycled consumables (2-hour SLA window).
3. **Multi-Compartment Dynamic Batching:** Each rover is equipped with 3 lockable modular cargo bins. The system decides whether to dispatch a rover immediately with a single urgent item or hold for 3–5 minutes to batch co-located packages heading to adjacent campus buildings (e.g., Central Receiving → Science Complex → Student Center).
4. **Dispatching & Resource Contention:** Allocates ready rovers from the active fleet. If all rovers are in transit or recharging, requests accumulate in building holding queues, triggering SLA violation penalties.
5. **Pedestrian-Interactive Transit & Bottlenecks:** Rovers navigate outdoors between campus buildings. Walkway transit times are *not static delays*; they fluctuate dynamically based on the 10-minute passing periods between classes, where pedestrian density surges by up to 500%, forcing rovers to decelerate, detour, or halt.
6. **Delivery Handshake & Unsuccessful Delivery Subsystem:** Upon arriving at a destination hub, the recipient is alerted. If the recipient does not claim the package within a 3-minute window:
   - *Primary Fallback:* Deposit package into a secure automated campus parcel locker (if slots are available).
   - *Secondary Fallback:* Abort delivery, flag as "Delivery Exception", retain package in rover cargo bin, and schedule a return trip to Central Receiving, directly restricting cargo capacity.
7. **Energy Replenishment & Depot Management:** Rovers monitor State-of-Charge (SOC). When entering charging depot (Station D), rovers contend for limited inductive charging slots (2 pads for 3–5 rovers). The system balances fast-charging downtime against peak dispatch availability.

1.2 Agent-Based Modeling (ABM): Agent Types, Attributes, States & Behaviors

Agent Type	Internal Attributes	Statechart & Operational States	Interactive Behaviors
Autonomous Courier Rover (Active Agent)	• ID, (X, Y) coordinates, velocity • Battery State of Charge (0–100%) • 3-Slot Modular Cargo Inventory • Cumulative mileage & degradation rate	• `IDLE_DEPOT` • `LOADING_BATCH` • `IN_TRANSIT` • `DELIVERY_HANDSHAKE` • `NAVIGATING_TO_CHARGER` • `FAST_CHARGING`	• Dynamic speed throttled by local pedestrian density. • Senses charging dock occupancy. • Communicates status to Central Dispatch Coordinator.
Pedestrian Student Crowds (Environmental Agent)	• Origin & destination building • Walking speed (1.0–1.4 m/s) • Class schedule timestamp	• `IN_CLASS` (Off-path) • `CLASS_PASSING_PERIOD` (On walkway) • `LOITERING` (Courtyard/plaza)	• Crosses pedestrian pathways in stochastic swarms during 10-minute class breaks. • Acts as dynamic obstacle fields that slow rover navigation.
Campus Building Hubs (Fixed Agent)	• Building name, Coordinates • Outbound parcel queue • Smart Locker Capacity (Occupied / Free)	• `NORMAL_LOAD` • `QUEUE_CONGESTED` • `LOCKER_FULL`	• Generates parcel orders. • Receives delivered packages. • Holds failed deliveries in automated lockers.
Central Charging Depot (Resource Agent)	• Number of Inductive Charge Pads (2) • Queue buffer capacity • Charge rate (0.8% SOC / min)	• `PADS_AVAILABLE` • `ALL_PADS_OCCUPIED` • `DEPOT_CONGESTED`	• Charges docked rovers. • Queues rovers with low battery. • Releases rovers upon reaching 90% SOC.

1.3 Causal-Loop Feedback Dynamics



Key Loops: *Balancing Loop B1* limits fleet utilization as rovers exhaust battery reserves and queue at Station D. *Reinforcing Loop R1* inflates delivery delays: when multiple rovers dock at once, remaining rovers face higher workload, depleting faster and escalating delivery times. *Balancing Loop B2* captures pedestrian walkway crowding: higher rover deployments create walkway interference, triggering speed throttling.

1.4 Reinforcement Learning: Sequential Multi-Objective Decision Making

The RL supervisory agent does not merely find the shortest path; it solves a complex, sequential **Multi-Objective Markov Decision Process (MOMDP)** coordinating batching, assignment, and preemptive energy management:

State Space Vector $S_t \in \mathbb{R}^{10}$ (Observed Every Decision Step):

1. `[Q_HubA, Q_HubB, Q_HubC]` : Queue depths at Central Mail, Science Lab, and Admin Hub.
2. `[U_HubA, U_HubB, U_HubC]` : Number of urgent (Tier 1) parcels awaiting immediate pickup.
3. `[SOC_R1, SOC_R2, SOC_R3]` : Battery percentage (0.0–1.0) of each rover in the fleet.
4. `Ped_Density` : Real-time pedestrian walkway congestion index (0.2 calm to 1.0 passing rush).
5. `Docks_Free` : Number of available charging pads at Station D (0, 1, or 2).

Action Space A_t (Discrete Combinatorial Decisions):

- `Action 0..2` : **Immediate Express Dispatch:** Assign available rover to Hub A, B, or C for single urgent delivery.
- `Action 3..5` : **Batch & Route:** Instruct rover to load 2–3 co-located parcels across adjacent buildings before departure.
- `Action 6` : **Opportunistic Top-off Charge:** Pull a moderately charged rover (e.g. 55% SOC) into Station D for a brief top-off before the upcoming class rush hour.
- `Action 7` : **Congestion-Bypass Route:** Route rover via secondary perimeter pathway to avoid high-density pedestrian courtyards.
- `Action 8` : **Standby:** Hold rover at strategic central node to conserve battery.

Multi-Objective Reward Function (R_t):

$$R_t = +15 \times N_{delivered} - 0.5 \times \sum WaitTimes - 25 \times \sum SLA_Violations - 0.8 \times EnergyConsumed - 100 \times I(Rover\ Stranded)$$

Comparing Q-Learning vs. PPO: Tabular/DQN Q-learning handles discrete assignment states well in low-traffic conditions, but **PPO (Proximal Policy Optimization)** excels at continuous policy smoothing—anticipating pedestrian crowd surges and scheduling opportunistic top-off charging *prior* to class breaks, preventing deadlocks and SLA violations.

SECTION 2: OPTION 1 — MDC AI & ROBOTICS LAB LOGISTICS &

TESTING OPTIMIZATION

For complete rigor, we also expand Option 1 as requested, detailing its multi-process scope within the MDC AI & Robotics facility:

- **Operational Processes:** Student bill-of-materials request → inventory shelf part retrieval (Station A) → soldering bench transport (Station B) → QA sensor calibration delay (Station C) → tool return → charging cycle (Station D). Bottlenecks occur when multiple capstone teams request testing bench access simultaneously while AMRs deplete batteries.
- **Agent Types:** 3 Differential AMRs (Inspector Alpha, Logistics Beta, Courier Gamma), Unitree G1 Humanoid (safety inspector), Student workbenches (A–C), Inductive charge pads (Station D).
- **Sequential RL Problem:** Balancing high student lab request throughput against battery lifetime and workbench tool contention using Q-Learning vs. PPO.

SECTION 3: 4-STUDENT TECHNICAL RESPONSIBILITIES & 4-WEEK ROADMAP

Team Member	Dedicated Technical Domain	Substantial Course Deliverables
Jorge A Mendoza	**AnyLogic Lead Architect**	• Builds multi-method AnyLogic `.alp` model. • Connects DES process pipelines (parcel generation, SLA queues, locker fallback). • Configures ABM rover statecharts, campus pedestrian paths, and 3D spatial animation canvas.
Jordan Gonzalez	**Reinforcement Learning Engineer**	• Implements Python Gymnasium environment with multi-objective reward formulation. • Trains and tunes **Q-Learning** policy. • Trains and evaluates **PPO** policy (Stable-Baselines3/PyTorch). • Exports policy weights, convergence curves, and comparative metrics.
Illia Shybanov	**Streamlit Dashboard & UI Specialist**	• Builds full 5-tab interactive Streamlit web application (`app.py`). • Integrates real-time parameter sliders (arrival rates, rover fleet count, charging pads). • Plots interactive Plotly KPI visuals (SLA compliance, throughput, energy burn). • Connects 3D twin telemetry preview and policy comparison graphs.
Ricardo Molina Gonzalez	**Experiments & Technical Reporting Lead**	• Executes the 5 required AnyLogic experiments: Optimization, Compare Runs, Monte Carlo (100 runs), Sensitivity Analysis, and Calibration. • Writes the 2–3 page academic report detailing non-obvious findings and recommendations. • Builds the 10–12 minute presentation slide deck. • Manages final package assembly (including individual Module 4 zips).

Conclusion & Requested Approval

Based on the professor's strong preference, **our team enthusiastically selects Option 2 (MDC Smart Campus Autonomous Courier Rover Logistics)** as our primary group project. The mathematical foundations, simulation architecture, code scaffolds, and team work breakdown are fully prepared for immediate execution.